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322752 (22)

BE (7th Semester)

Examination, April-May, 2014

Branch : CSE, IT

ADVANCED COMPUTER ARCHITECTURE

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) of all question is compulsory.

(ii) Solve any two parts out of 'b' 'c' & 'd' of question.

Q. 1. (a) Pipelining is one way a implement parallelism. Comment & justify. 2

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(b) With reference to linear pipelining define following term. Give mathematical expression :

(i) Clock period

(ii) Speed up

(iii) Efficiency

(iv) Throughput

(c) Design a floating point adder in form of pipeline system having four stages. Take one example and explain the process. 1

(d) With reference to VLIW system, give technical reason, how instruction level parallelism differs with VLIW. Also elaborate on process of VLIW system. 1

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Q. 2. (a) What do you mean by multilevel cache system ? 2

(b) What is the main reason of difference in bandwidth mismatch ? Give atleast two technique for balancing bandwidth. Give suitable diagrams & mathematical illustrations. 7

(c) Give functional block diagram of memory system, having following features : 7

(i) No. of memory modules are 2^m

(ii) All modules are of same size

(iii) One module can store $n-m$ number of words

(iv) Address bits are = n

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(d) What do you mean by cache coherence Give two different techniques to hanc coherence problem.

Q. 3. (a) Explain why data flow computers are les popular.

(b) Take two matrix of 2 × 2 dimension & explai program decomposition for stati multiprocessor. 7

(c) Write short note on controll flow computer & data flow computers. 7

(d) For a program given below, give data flow graph and find out time required for sequential execution Assume that we have 4 processor working on data flow concept, now recalculate execution time 7

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4. (a) What do you mean by vector processing ? 2

(b) Give architectural block diagram of simple vector super computer. Explain its working. 7

(c) Give functional block diagram of SIMD computer. Take one example to sum first 100 natural number. Develop a parallel algorithm for same. 7

(d) Proof that the best parallel algorithm written for an n-processor EREW-PRAM model can be no more than O(logn) times slower than any algo for a CRCW model of PRAM having the same number of processor. 7

5. (a) Why mesh structure is versatile one. Explain

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(b) List out and explain three popular interconnection networks.

(c) Give details of butterfly interconne network, with its application.

(d) Give short note on multistage and combir networks.

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